1. Derive, look up, guess, borrow, or steal the Hamiltonian operator **H**.

2. Prepare an initial state $|\Psi(0)\rangle$.

3. Find the eigenvalues and eigenvectors of **H** by solving the time-independent Schrödinger equation,

$$\mathbf{H}|E_j\rangle = E_j|E_j\rangle.$$

4. Use the initial state-vector $|\Psi(0)\rangle$, along with the eigenvectors $|E_j\rangle$ from step 3, to calculate the initial coefficients $\alpha_j(0)$:

$$\alpha_j(0) = \langle E_j|\Psi(0)\rangle.$$

5. Rewrite $|\Psi(0)\rangle$ in terms of the eigenvectors $|E_j\rangle$ and the initial coefficients $\alpha_j(0)$:

$$|\Psi(0)\rangle = \sum_j \alpha_j(0)|E_j\rangle.$$

> What we've done so far is to expand the initial state-vector $|\Psi(0)\rangle$ in terms of the eigenvectors $|E_j\rangle$ of **H**. Why is that basis better than any other? Because **H** *tells us how things evolve with time*. We will use that knowledge now.

6. In the above equation, replace each $\alpha_j(0)$ with $\alpha_j(t)$ to capture its time-dependence. As a result, $|\Psi(0)\rangle$ becomes $|\Psi(t)\rangle$: